FORGED BY FREEDOM

BAM-15

BAM-15 — Mitochondrial Protonophore Uncoupler

FAT LOSS	MITOCHONDRIAL	UNCOUPLER	THERMOGENIC	EMERGING

WHAT IS IT?

BAM-15 is a next-generation mitochondrial protonophore — a mitochondrial uncoupling agent that 'short-circuits' the electron transport chain, causing energy (calories) to be released as heat instead of being stored as ATP. Unlike DNP (2,4-dinitrophenol), BAM-15 acts only on the inner mitochondrial membrane and does not dissipate the plasma membrane electrochemical gradient — making it substantially safer in preclinical models. Published in Nature Communications (2020) — significant fat loss without diet change in mice.

DOSING AT A GLANCE

No human trials	Preclinical dose	Likely route	Status
Unknown (human)	Mouse: 50 mg/kg	Oral	Research only

KEY BENEFITS

- Significant fat loss by converting stored energy to heat (thermogenesis)
- Preserves lean muscle mass during fat loss — unlike most caloric-restriction approaches
- Improves insulin sensitivity — reduces insulin resistance in obese mouse models
- Reduces liver fat (hepatic steatosis) and systemic inflammation
- No suppression of appetite — mechanism is purely thermogenic
- Improved mitochondrial efficiency markers
- Synergistic with exercise — enhances the uncoupling that occurs naturally during training
- Unlike DNP: does not raise core body temperature excessively (safer thermodynamic profile)

HOW IT WORKS

BAM-15 is a proton carrier — it shuttles protons (H+) across the inner mitochondrial membrane down the electrochemical gradient, bypassing ATP synthase. This 'uncouples' oxidative phosphorylation from ATP production, so the energy from fuel oxidation is released as heat. The cell compensates by burning more fuel — creating a caloric deficit at rest. BAM-15 specificity for the mitochondrial membrane prevents the dangerous plasma membrane depolarization that makes DNP lethal at higher doses.

PROTOCOLS

- **CRITICAL:** No human dose established — BAM-15 is pure preclinical research compound.
- **Animal studies:** 50 mg/kg/day oral in mice — human equivalent roughly 4 mg/kg (HED).
- **Current status:** Not available from mainstream peptide vendors; strictly research compound.
- **If experimenting:** Cardiovascular monitoring (body temp, heart rate, blood pressure) is essential.
- **DNP comparison:** BAM-15 appears safer in animals but do NOT extrapolate DNP dosing.

SIDE EFFECTS

- Unknown in humans — no human safety data exists
- Preclinical: no significant body temperature elevation at study doses (unlike DNP)
- Theoretical: hyperthermia, tachycardia, sweating at high doses (uncoupler class effect)
- Theoretical: mitochondrial dysfunction if mechanism is overwhelmed
- DNP class risk: uncouplers at toxic doses cause fatal hyperthermia — extreme caution required

STORAGE

Research chemical — store per supplier instructions. Typically -20°C freezer, sealed.

■ ■ ***EXTREME CAUTION — Research compound with zero human safety data. DNP, a related uncoupler, has caused multiple deaths. Do not extrapolate DNP experience to BAM-15. For research monitoring only.***